

# JPE Replication Report

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## 1 Paper Identification

|           |                                                                                    |
|-----------|------------------------------------------------------------------------------------|
| ID        | JPE-Richert-20240555                                                               |
| Author    | Richert, Eric                                                                      |
| Title     | Quantity Commitments in Multiunit Auctions:<br>Evidence from Credit Event Auctions |
| Iteration | 2                                                                                  |

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

## 2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. We found a substantial number of discrepancies. Please comment on each of those. Some could be tables which were not updated, but others seems very far off.

## 3 General

- ☒ Data Availability and Provenance Statements
  - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - ☐ License for Data (optional, but recommended)
  - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
  - ☒ Software Requirements
  - ☒ Controlled Randomness (as necessary)
  - ☐ Memory, Runtime, and Storage Requirements

- ☒ Description of programs/code
  - ☐ License for Code (Optional, but recommended)
- ☒ {REPLICATOR} to Replicators
  - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

## 4 High-level Description of Research Project

- ☐ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
  - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
  - ☐ A PDF copy of IRB approval is contained in the package.
  - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
  - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
  - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
  - ☐ A PDF copy of IRB approval is contained in the package.
  - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

## 5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

## 5.1 Data Sources

| file/name              | public | private | DOI/URL | Access described | cited readme | cited paper |
|------------------------|--------|---------|---------|------------------|--------------|-------------|
| immspreads_clean.csv   | no     | yes     | yes     | yes              | yes          | yes         |
| openinterest_clean.csv | no     | yes     | yes     | yes              | yes          | yes         |
| limitorders_clean.csv  | no     | yes     | yes     | yes              | yes          | yes         |
| auctionlistid.csv      | no     | yes     | yes     | yes              | yes          | yes         |
| bondtypes.csv          | no     | yes     | yes     | yes              | yes          | yes         |
| gosyop23q6grk5y1.csv   | no     | yes     | yes     | yes              | yes          | yes         |

## 6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

### 6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
  - MATLAB R2023a
  - Toolboxes: Statistics and Machine Learning Toolbox, Optimization Toolbox, Parallel Computing Toolbox
- ☐ Computational Requirements specified as follows:
  - Hardware: 20-core cluster node or a standard multi-core desktop
  - Storage: Sufficient space for large intermediate .mat files (up to ~700 MB for smc\_cfs\_np.mat)
  - **Storage actually requires ~56 GB**
- ☒ Time Requirements specified as follows:
  - approximately 20 hours on a 20-core cluster node, or 3–4 days on a desktop
- ☒ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

## 7 Analyzing Data and Code Files

### 7.1 Potential Personal Identifiable Information (PII)

△ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

**Summary:** - Data files with PII indicators: 5 - Variables flagged in data: 6 - Code files with PII references: 54 - PII references in code: 610

#### 7.1.1 Summary of Flagged Files

| File Type | File                                     | Variables/References | PII Categories            |
|-----------|------------------------------------------|----------------------|---------------------------|
| Data      | <code>auctionlistid.csv</code>           | 1                    | name                      |
| Data      | <code>bondtypes.csv</code>               | 2                    | name, loc                 |
| Data      | <code>immspreads_clean.csv</code>        | 1                    | name                      |
| Data      | <code>limitorders_clean.csv</code>       | 1                    | name                      |
| Data      | <code>openinterest_clean.csv</code>      | 1                    | name                      |
| Code      | <code>BsplineBasis3.m</code>             | 24                   | city, second, son, degree |
| Code      | <code>BsplineEval3.m</code>              | 1                    | second                    |
| Code      | <code>acrossrounds.m</code>              | 1                    | second                    |
| Code      | <code>basic_estimator.m</code>           | 56                   | lon, lat, city            |
| Code      | <code>basic_estimator_copypdrop.m</code> | 56                   | lon, lat, city            |
| Code      | <code>bondpriceimport.m</code>           | 14                   | name, loc                 |
| Code      | <code>bondprices.m</code>                | 1                    | lat                       |
| Code      | <code>build_bondforestimation.m</code>   | 2                    | name                      |

| File Type | File                                               | Variables/References | PII Categories                   |
|-----------|----------------------------------------------------|----------------------|----------------------------------|
| Code      | <code>calculate_surplusraw.m</code>                | 5                    | lon, son                         |
| Code      | <code>cdf_estimator.m</code>                       | 1                    | lat                              |
| Code      | <code>cdf_estimator_imm.m</code>                   | 2                    | lat                              |
| Code      | <code>complex_bootstrap.m</code>                   | 6                    | second, lon, lat                 |
| Code      | <code>complex_estimator_step1.m</code>             | 71                   | lon, lat, city                   |
| Code      | <code>complex_estimator_step2.m</code>             | 12                   | lon, lat                         |
| Code      | <code>complex_estimator_step2b.m</code>            | 11                   | lon, lat                         |
| Code      | <code>data_summary.m</code>                        | 21                   | name, block, loc, lat, second    |
| Code      | <code>debug_bondforeestimation.m</code>            | 3                    | name                             |
| Code      | <code>doubleauction0uter_1stepspecial_yin.m</code> | 3                    | lat                              |
| Code      | <code>dscatter.m</code>                            | 11                   | loc, location, lon, name, second |
| Code      | <code>emcdf.m</code>                               | 1                    | lat                              |
| Code      | <code>firststagebidding.m</code>                   | 10                   | loc, lat                         |
| Code      | <code>heatscatter.m</code>                         | 9                    | name, lat                        |
| Code      | <code>jacobianest.m</code>                         | 13                   | loc, location, lat, second       |
| Code      | <code>job0_validate.m</code>                       | 5                    | name, loc                        |
| Code      | <code>job0_validate.sh</code>                      | 1                    | name                             |
| Code      | <code>job1_dataprep.m</code>                       | 1                    | name                             |
| Code      | <code>job1_dataprep.sh</code>                      | 1                    | name                             |
| Code      | <code>job2a_estimation.m</code>                    | 3                    | loc, location, name              |
| Code      | <code>job2a_estimation.sh</code>                   | 1                    | name                             |
| Code      | <code>job2b_estimation.m</code>                    | 3                    | loc, location, name              |

| File Type | File                                    | Variables/References | PII Categories           |
|-----------|-----------------------------------------|----------------------|--------------------------|
| Code      | <code>job2b_estimation.sh</code>        | 1                    | name                     |
| Code      | <code>job3a_postcfs.m</code>            | 4                    | loc, location, lat, name |
| Code      | <code>job3a_postcfs.sh</code>           | 1                    | name                     |
| Code      | <code>job3b_postcfs.m</code>            | 3                    | loc, location, name      |
| Code      | <code>job3b_postcfs.sh</code>           | 1                    | name                     |
| Code      | <code>job3c_postcfs.m</code>            | 3                    | loc, location, name      |
| Code      | <code>job3c_postcfs.sh</code>           | 1                    | name                     |
| Code      | <code>main_cds.m</code>                 | 3                    | lat, loc, location, name |
| Code      | <code>normalize_prices.m</code>         | 28                   | lon, lat, second, name   |
| Code      | <code>npestimator.m</code>              | 20                   | lon                      |
| Code      | <code>permn.m</code>                    | 4                    | son, name, lat, second   |
| Code      | <code>pmcalculation.m</code>            | 1                    | lat                      |
| Code      | <code>postestimation_clean.m</code>     | 14                   | lat, lon, son, url       |
| Code      | <code>postmain_cfs.m</code>             | 1                    | lat                      |
| Code      | <code>pricechangebs.m</code>            | 1                    | loc, location            |
| Code      | <code>robustnesschecks.m</code>         | 6                    | lat, lon                 |
| Code      | <code>round1_quotescalibration.m</code> | 8                    | lat, lon                 |
| Code      | <code>smc_cfs_yin.m</code>              | 30                   | lat, block, loc          |
| Code      | <code>table2latex.m</code>              | 20                   | lat, name, son           |
| Code      | <code>truthfulpimmcheck.m</code>        | 3                    | lat                      |
| Code      | <code>v_correction.m</code>             | 3                    | lon                      |
| Code      | <code>weightedMedian.m</code>           | 4                    | lat                      |

| File Type | File                                                | Variables/References | PII Categories                  |
|-----------|-----------------------------------------------------|----------------------|---------------------------------|
| Code      | <code>weightedcorrs.m</code>                        | 100                  | lat, son, second, loc, location |
| Code      | <code>weightsolnpricespartialgridINTs1_yin.m</code> | 1                    | lon                             |

See [Appendix](#) for detailed listing of all flagged instances.

## 7.1.2 File Paths Report

Generated on 2026-04-23T12:20:21.284

**Warning:** Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for all `STATA` applications.)

| Files Analyzed | Windows Paths | Unix Paths | Mixed Paths | Drive Letters ( <code>C:\</code> etc) |
|----------------|---------------|------------|-------------|---------------------------------------|
| 80             | 10            | 29         | 15          | 0                                     |

## 7.1.3 Duplicate Files Report

We did not find any duplicate files.

## 7.1.4 Zero Size Files Report

We did not find any zero sized files.

## 7.1.5 Large Files Report

We found the following files larger than 100MB:

| Filename                                                        | Size (MB) |
|-----------------------------------------------------------------|-----------|
| /CDS/confidential-data-not-for-publication/gosyop23q6grk5y1.csv | 129.31    |

## 7.2 Code description

There are provided 37 MATLAB, 1 of which is the master script ( `code/main_cds.m` ).

| cloc         |       | github.com/AIDanial/cloc v 2.02 T=0.13 s (666.3 files/s, 202312.1 lines/s) |         |       |
|--------------|-------|----------------------------------------------------------------------------|---------|-------|
| Language     | files | blank                                                                      | comment | code  |
| CSV          | 5     | 0                                                                          | 0       | 14416 |
| MATLAB       | 73    | 1152                                                                       | 2113    | 7771  |
| Markdown     | 1     | 152                                                                        | 0       | 319   |
| Bourne Shell | 7     | 22                                                                         | 98      | 69    |
| ---          | ---   | ---                                                                        | ---     | ---   |
| SUM:         | 86    | 1326                                                                       | 2211    | 22575 |

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

## 7.3 Computing Environment of the Replicator

Hardware:

- Nuvolos: Ubuntu 24.04.1 LTS, 16 vCPU + 16 GB RAM

Software:

- Matlab R2023a with Intel compilers

## 8 Replication

- Downloaded code and data from dropbox.
- Copied git repo and unzipped the replication package into this repo.
- Set up the computing environment on Nuvolos and scaled the hardware to 16 vCPU and 64 GB RAM.
- Modified `code/main_cds.m` to set `ncores = 16` to match the available cluster hardware as instructed in the `README.md`
- Set working directory and ran the code with MatLab 2023a as per `README.md`
- Committed changes

## 9 Findings



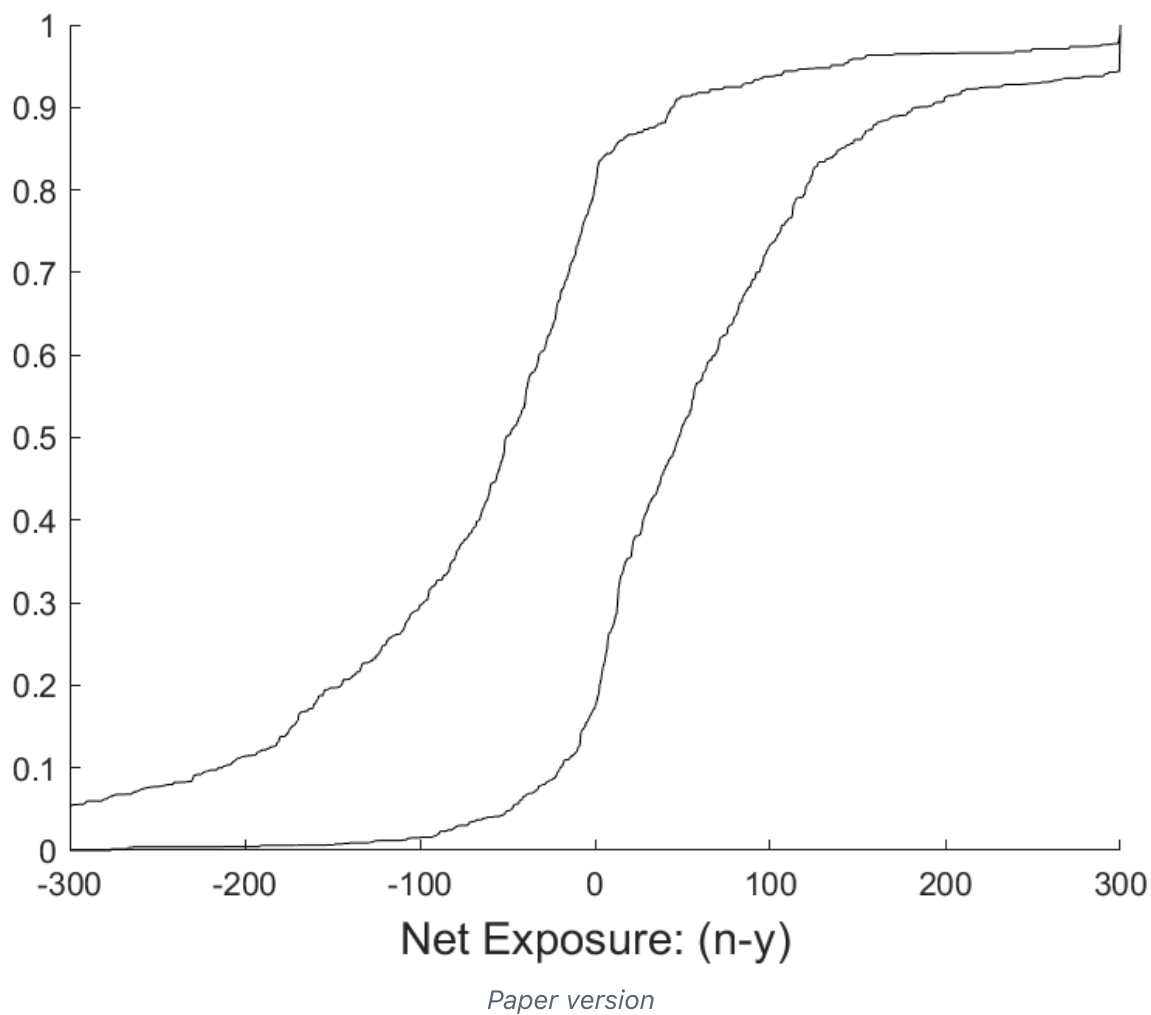
## 9.1 Data Preparation Code

## 9.2 Tables and Figures

| Figure/Table #    | Program                | Line Number | Reproduced? |
|-------------------|------------------------|-------------|-------------|
| Figure 1A         | normalize_prices.m     | 458         | Yes         |
| Figure 1B         | normalize_prices.m     | 469         | Yes         |
| Figure 3          | /                      |             | Yes         |
| Figure 4 (left)   | postestimation_clean.m | 395         | No          |
| Figure 4 (middle) | postestimation_clean.m | 408         | No          |
| Figure 4 (right)  | postestimation_clean.m | 416         | No          |
| Table 1           | data_summary.m         | 130         | Yes         |
| Table 2           | data_summary.m         | 158         | No          |
| Table 3           | normalize_prices.m     | 263         | No          |
| Table 4           | postestimation_clean.m | 219         | No          |
| Table 5           | postmain_cfs.m         | 76          | No          |
| Table A.1         | normalize_prices.m     | 437         | No          |
| Table OS.1        | data_summary.m         | 77          | No          |
| Table OS.2        | data_summary.m         | 305         | No          |
| Table OS.3        | data_summary.m         | 200         | No          |
| Table OS.4        | normalize_prices.m     | 326         | No          |
| Table OS.5        | normalize_prices.m     | 379         | No          |
| Table OS.6        | main_cds.m             | 153         | No          |
| Figure OS.3       | bondprices.m           | 69          | Yes         |
| Figure OS.4       | data_summary.m         | 356         | Yes         |
| Figure OS.5       | data_summary.m         | 334         | Yes         |
| Figure OS.6       | robustnesschecks.m     | 25, 31      | No          |

| Figure/Table # | Program                | Line Number | Reproduced? |
|----------------|------------------------|-------------|-------------|
| Figure OS.7    | firststagebidding.m    | 97          | Yes         |
| Figure OS.8    | postestimation_clean.m | 330         | No          |

- Figure 4 (left): lower line is different



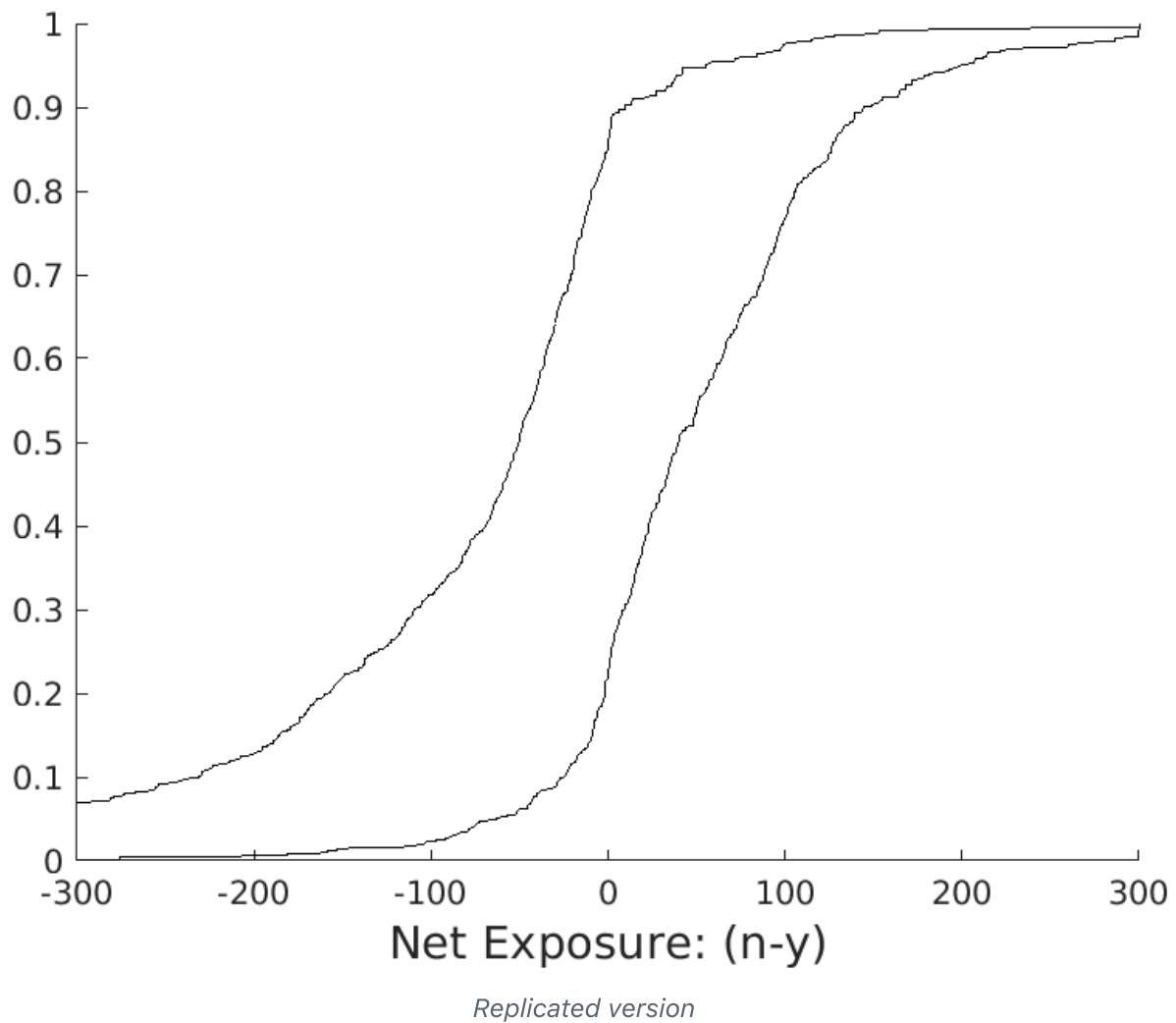
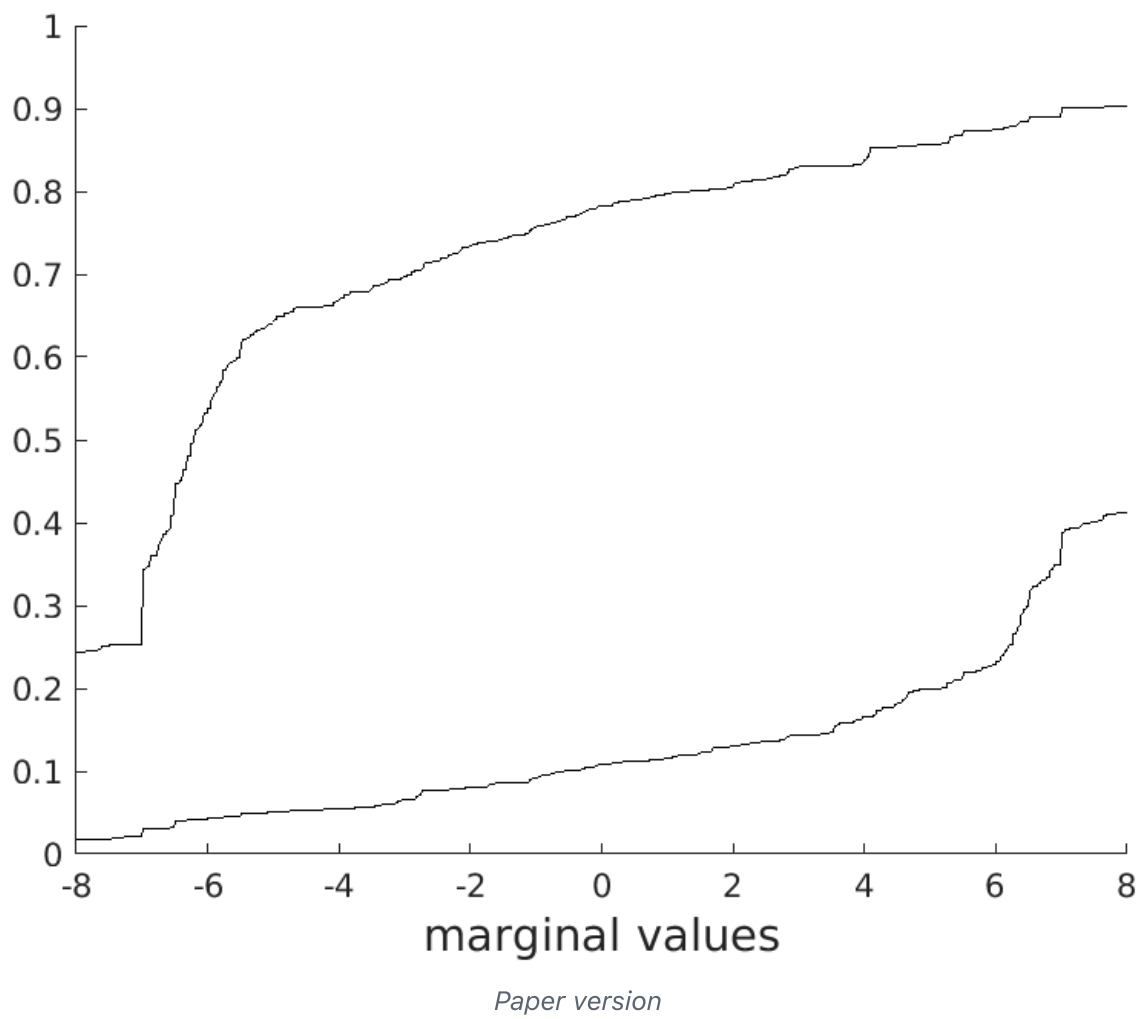


Figure 1: Figure 4 (left)

- Figure 4 (center): upper line is different



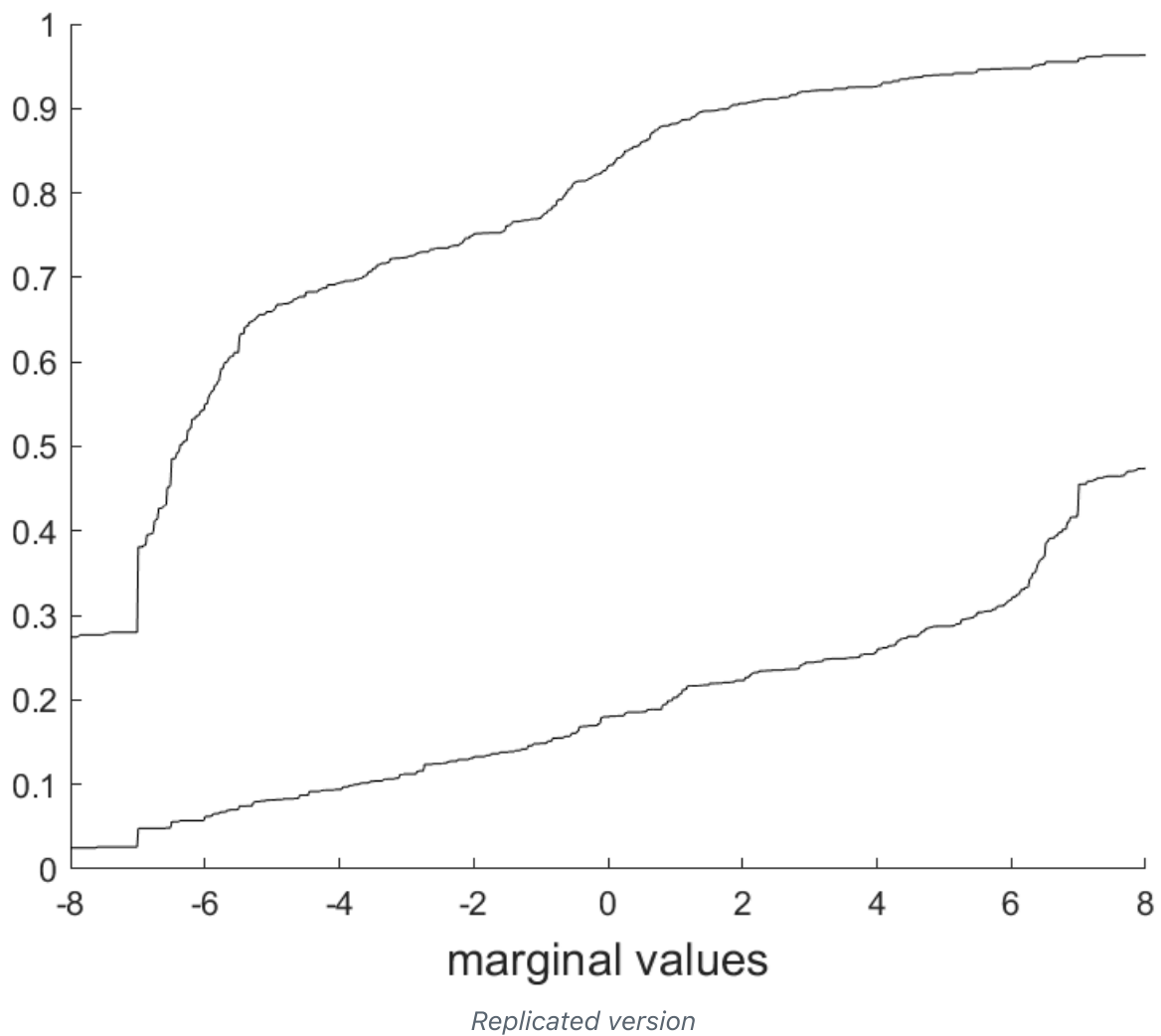
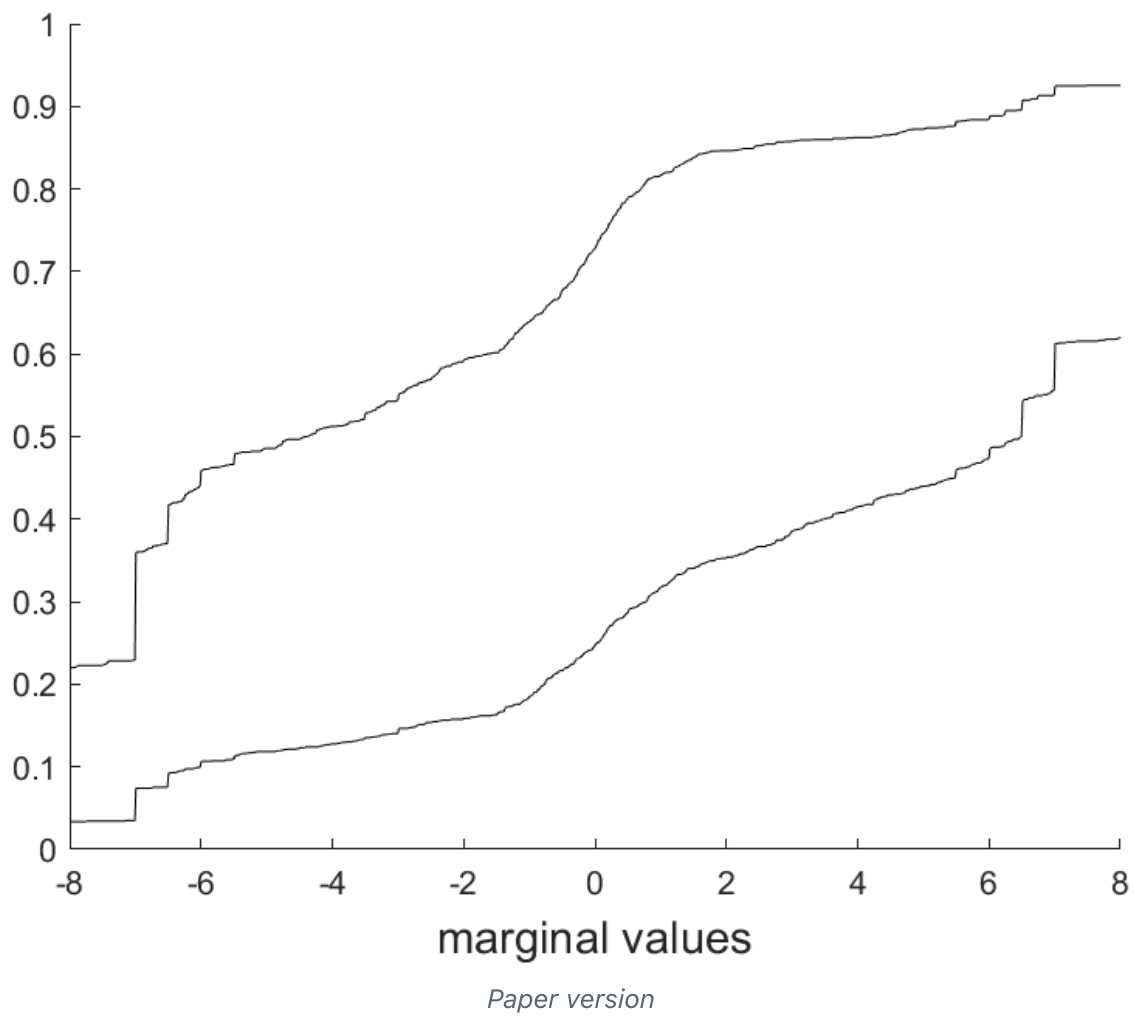


Figure 2: Figure 4 (center)

- Figure 4 (right): both lines are different



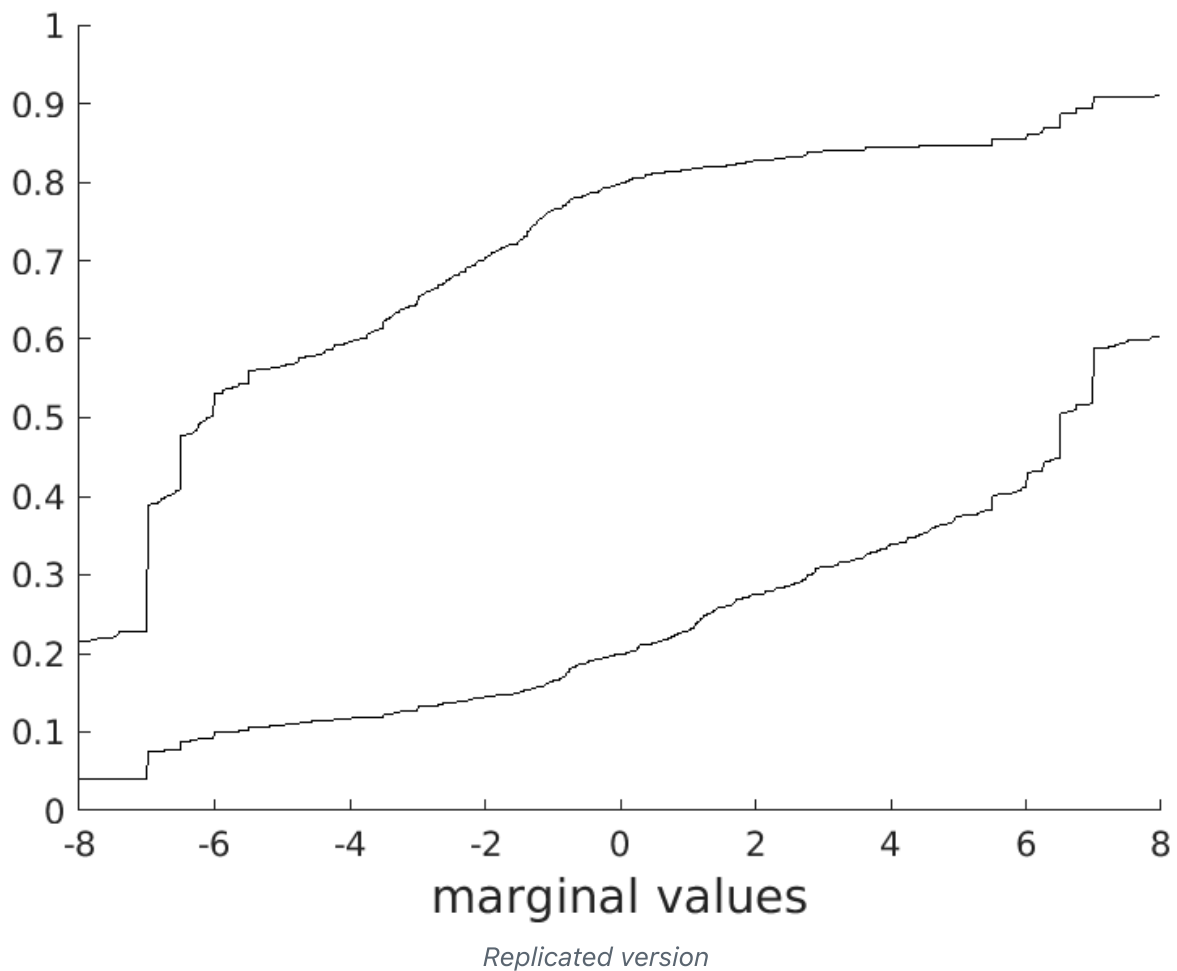


Figure 3: Figure 4 (right)

- Table 2: estimates do not match

| Name                                      | Submissions |      |       | Sorted Bids |      | Sorted Offers |       |
|-------------------------------------------|-------------|------|-------|-------------|------|---------------|-------|
|                                           | ID          | Bid  | Offer | ID          | Bid  | ID            | Offer |
| Barclays Bank PLC                         | 1           | 53   | 58    | 5           | 56   | 8             | 55    |
| BNP Paribas SA                            | 2           | 54   | 59    | 2           | 54   | 1             | 58    |
| Credit Suisse                             | 3           | 53   | 58    | 4           | 53.5 | 3             | 58    |
| Deutsche Bank                             | 4           | 53.5 | 58.5  | 9           | 53.5 | 6             | 58    |
| Goldman Sachs International               | 5           | 56   | 61    | 3           | 53   | 7             | 58    |
| JP Morgan Securities LLC                  | 6           | 53   | 58    | 6           | 53   | 4             | 58.5  |
| Merrill Lynch, Pierce, Fenner & Smith Inc | 7           | 53   | 58    | 7           | 53   | 9             | 58.5  |
| Morgan Stanley & Co., LLC                 | 8           | 50   | 55    | 1           | 53   | 2             | 59    |
| Societe Generale                          | 3           | 53.5 | 58.5  | 8           | 50   | 5             | 61    |

Paper version

Table 2: Initial Stage Price Quotes (Parker Drilling)

| Dealer          | Bid   | Offer |
|-----------------|-------|-------|
| bnpparibas      | 54.00 | 59.00 |
| barclays        | 53.00 | 58.00 |
| societegenerale | 53.50 | 58.50 |
| merrilllynch    | 53.00 | 58.00 |
| goldmansachs    | 55.00 | 60.00 |
| morganstanley   | 51.00 | 56.00 |
| deutschebank    | 53.50 | 58.50 |
| jpmorgan        | 53.00 | 58.00 |
| suisse          | 53.00 | 58.00 |
| IMM = 55.75     |       |       |

*Replicated version*

Figure 4: Table 2

- Table 3: estimates do not match

|                                                                     | Mean   | Sd     | [P10,P90]    |
|---------------------------------------------------------------------|--------|--------|--------------|
| Number of Dealers                                                   | 10.97  | 2.62   | [8,14]       |
| Price (\$.01)                                                       | 43.81  | 33.20  | [4.18,93.93] |
| IMM (\$.01)                                                         | 44.07  | 32.76  | [4.40,91.78] |
| Within-Auction deviation in Quotes (\$.01)                          | 1.47   | 1.28   | [0.41,2.75]  |
| NOI  (\$millions)                                                   | 120.01 | 382.90 | [0.29,235.2] |
| Total Commitments $\sum  \text{initial commitment}_i $ (\$millions) | 224.4  | 568.5  | [0.44,554.6] |
| Auction Coverage (shares)                                           | 9.71   | 34.38  | [1,12.09]    |
| Individual Shares of Total Commitment                               | 0.09   | 0.19   | [0,0.30]     |
| Maximum Quantity Stage 2 (shares)                                   | 0.43   | 0.70   | [0, 1.15]    |

*Paper version*



Table 3: Auction Description

|                                       | Mean   | Sd     | $[P_{10}, P_{90}]$ |
|---------------------------------------|--------|--------|--------------------|
| Number of Dealers                     | 10.97  | 2.62   | [8.00, 14.00]      |
| Price (                               |        |        |                    |
| .01)                                  | 43.81  | 33.20  | [4.19, 93.93]      |
| IMM (                                 |        |        |                    |
| .01)                                  | 44.07  | 32.76  | [4.40, 91.78]      |
| Within-Auction deviation in Quotes (  |        |        |                    |
| .01)                                  | 1.48   | 1.28   | [0.42, 2.76]       |
| NOI  (                                |        |        |                    |
| millions)                             | 120.01 | 382.90 | [0.29, 235.16]     |
| Total Commitments                     |        |        |                    |
| sum $ y_i $ (                         |        |        |                    |
| millions)                             | 224.42 | 568.53 | [0.44, 554.59]     |
| Auction Coverage (shares)             | 9.43   | 34.39  | [1.00, 11.71]      |
| Individual Shares of Total Commitment | 0.09   | 0.19   | [0.00, 0.30]       |
| Maximum Quantity Stage 2 (shares)     | 0.42   | 0.68   | [0.00, 1.09]       |

$N = 198$  auctions (rows 1–7), 2172 bidder-level observations (rows 8–9).

*Replicated version*

Figure 5: Table 3

- Table 4: estimates do not match

|                         | Mean           | Variance       | Cov. Auction Price |
|-------------------------|----------------|----------------|--------------------|
| Auction Price           | 43.81          | 1102           | 1102               |
| Truthful Bidding Price  | [44.74, 52.42] | [783.47, 1235] | [912, 1179]        |
| Initial Market Price    | 44.07          | 1073           | 1080               |
| Default Event Indicator | 0.02           | 0.02           | -                  |

*Paper version*

Table 4: Statistics to Evaluate Auction Performance

|                         | Mean           | Variance       | Cov. Auction Price |
|-------------------------|----------------|----------------|--------------------|
| Auction Price           | 43.81          | 1102           | 1102               |
| Truthful Bidding Price  | [44.74, 52.41] | [784.81, 1280] | [913, 1179]        |
| Initial Market Price    | 44.07          | 1073           | 1080               |
| Default Event Indicator | 0.02           | 0.02           | —                  |

*Replicated version*

Figure 6: Table 4

- Table 5: estimates do not match

ions are presented in the Table.

|                      | Mean           | Variance     | $Cov(P^c, P^v)$ | Surplus    |
|----------------------|----------------|--------------|-----------------|------------|
| Double Auction Price | [44.45, 44.98] | [1150, 1247] | [1241, 1292]    | [0.6, 3.2] |

Paper version

Table 5: Change in Auction Format

|                      | Mean           | Variance     | $Cov(P^c, P^v)$ | Surplus    |
|----------------------|----------------|--------------|-----------------|------------|
| Double Auction Price | [44.17, 44.83] | [1186, 1321] | [1241, 1309]    | [0.8, 2.3] |

Replicated version

Figure 7: Table 5

- Table A1: estimates do not match

bidding resulting from optimal bidding in an CIP v model. Standard errors in parentheses.

|          | Slope of bids     |               | Mean bid<br>(NOI<0) | Mean bid<br>(NOI>0) |           | Mean bid       |
|----------|-------------------|---------------|---------------------|---------------------|-----------|----------------|
| IMM var. | -0.009<br>(0.035) | $p_i^M - p^M$ | -0.70<br>(0.444)    | -0.49<br>(1.24)     | Mean Opp. | 1.13<br>(1.52) |

Paper version

Table 6: Evidence of Independent Private Values

|               | Slope of bids    | Mean bid         |                 | Mean bid       |
|---------------|------------------|------------------|-----------------|----------------|
|               |                  | NOI<0            | NOI>0           |                |
| IMM var.      | 0.022<br>(0.150) |                  |                 |                |
| $p_i^M - p^M$ |                  | -0.70<br>(1.245) | -0.49<br>(1.13) |                |
| Mean Opp.     |                  |                  |                 | 1.13<br>(1.52) |
| $N$           | 2172             | 379              | 550             | 2172           |

Standard errors in parentheses. Each column includes additional controls (see text).

Replicated version

Figure 8: Table A1

- Table OS1: estimates do not match

|                      | Mean  | Sd    | [P10,P90] |
|----------------------|-------|-------|-----------|
| N Bonds              | 10.77 | 27.70 | [1,16.90] |
| Max maturity (years) | 10.42 | 11.93 | [0,26.9]  |
| Min maturity (years) | 2.97  | 4.92  | [0.6,90]  |
| Max coupon %         | 5.5   | 4.8   | [0,10.75] |
| Min coupon %         | 2.7   | 3.3   | [0,7.62]  |
| FRN %                | 33.8  | 41.7  | [0,100]   |

*Paper version*

Table 3: Eligible Bonds Description

|                      | Mean  | Sd    | [P <sub>10</sub> , P <sub>90</sub> ] |
|----------------------|-------|-------|--------------------------------------|
| N Bonds              | 10.77 | 27.70 | [1.00, 16.90]                        |
| Max maturity (years) | 10.42 | 11.93 | [2.00, 26.90]                        |
| Min maturity (years) | 2.97  | 4.91  | [0.00, 6.90]                         |
| Max coupon           |       |       |                                      |
| Min coupon           |       |       |                                      |
| FRN                  |       |       |                                      |

$N = 186$  auctions.

*Replicated version*

Figure 9: Table OS1

- Table OS2: estimates do not match

TABLE OS.2. Auction Participation

The following table presents summary statistics for participation of the bidders. Each number is a count of the number of auctions in which the bidder participated, submitted a positive or negative first stage quantity commitment or used each number of steps in their second stage bids.

| Bidder | Participated | $y_i > 0$ | $y_i < 0$ | 1 step | 2 steps | 3 steps | 4 steps | 5+ steps |
|--------|--------------|-----------|-----------|--------|---------|---------|---------|----------|
| 1      | 1            | 1         | 0         | 0      | 0       | 0       | 1       | 0        |
| 2      | 123          | 40        | 25        | 36     | 28      | 19      | 8       | 24       |
| 3      | 175          | 51        | 28        | 57     | 25      | 8       | 12      | 56       |
| 4      | 4            | 2         | 1         | 1      | 2       | 1       | 0       | 0        |
| 5      | 135          | 42        | 18        | 81     | 17      | 8       | 8       | 17       |
| 6      | 169          | 35        | 29        | 68     | 21      | 13      | 8       | 51       |
| 7      | 171          | 43        | 23        | 64     | 42      | 18      | 8       | 27       |
| 8      | 12           | 8         | 0         | 9      | 1       | 0       | 1       | 0        |
| 9      | 179          | 47        | 25        | 38     | 46      | 17      | 17      | 50       |
| 10     | 47           | 10        | 17        | 35     | 6       | 1       | 0       | 4        |
| 11     | 1            | 1         | 0         | 1      | 0       | 0       | 0       | 0        |
| 12     | 183          | 55        | 24        | 51     | 40      | 26      | 11      | 47       |
| 13     | 4            | 3         | 0         | 1      | 2       | 1       | 0       | 0        |
| 14     | 68           | 15        | 13        | 35     | 15      | 6       | 1       | 8        |
| 15     | 2            | 1         | 1         | 2      | 0       | 0       | 0       | 0        |
| 16     | 5            | 3         | 1         | 5      | 0       | 0       | 0       | 0        |
| 17     | 1            | 0         | 0         | 0      | 0       | 0       | 0       | 0        |
| 18     | 180          | 46        | 36        | 53     | 29      | 25      | 14      | 50       |
| 19     | 78           | 15        | 19        | 50     | 7       | 2       | 2       | 14       |
| 20     | 102          | 22        | 13        | 50     | 22      | 6       | 7       | 12       |
| 21     | 96           | 19        | 11        | 83     | 4       | 0       | 1       | 6        |
| 22     | 1            | 0         | 1         | 0      | 0       | 0       | 0       | 1        |
| 23     | 180          | 43        | 42        | 72     | 31      | 16      | 13      | 40       |
| 24     | 2            | 1         | 0         | 2      | 0       | 0       | 0       | 0        |
| 25     | 128          | 32        | 27        | 67     | 22      | 7       | 4       | 24       |

*Paper version*

Table 8: Auction Participation

| Bidder                                 | Participated | $y_i > 0$ | $y_i < 0$ | 1 step | 2 steps | 3 steps | 4 steps | 5+ steps |
|----------------------------------------|--------------|-----------|-----------|--------|---------|---------|---------|----------|
| abmrobanknv                            | 1            | 1         | 0         | 0      | 0       | 0       | 1       | 0        |
| bankofamerica                          | 131          | 41        | 25        | 38     | 28      | 19      | 8       | 27       |
| barclays                               | 187          | 52        | 28        | 61     | 25      | 8       | 12      | 57       |
| bearstearns                            | 4            | 2         | 1         | 1      | 2       | 1       | 0       | 0        |
| bnpparibas                             | 143          | 44        | 18        | 81     | 21      | 8       | 9       | 17       |
| citigroup                              | 178          | 36        | 29        | 70     | 23      | 13      | 8       | 52       |
| deutschebank                           | 183          | 46        | 23        | 66     | 42      | 18      | 8       | 28       |
| dresdner                               | 13           | 9         | 0         | 9      | 1       | 0       | 1       | 1        |
| goldmansachs                           | 190          | 48        | 26        | 38     | 47      | 18      | 17      | 53       |
| hsbc                                   | 52           | 11        | 21        | 36     | 10      | 1       | 0       | 4        |
| ingbanknv                              | 1            | 1         | 0         | 1      | 0       | 0       | 0       | 0        |
| jpmorgan                               | 193          | 56        | 27        | 53     | 40      | 26      | 11      | 50       |
| lehmanbrothers                         | 4            | 3         | 0         | 1      | 2       | 1       | 0       | 0        |
| merrillynch                            | 75           | 16        | 17        | 35     | 15      | 7       | 2       | 11       |
| mitsubishiufj                          | 2            | 1         | 1         | 2      | 0       | 0       | 0       | 0        |
| mizuho                                 | 5            | 3         | 1         | 5      | 0       | 0       | 0       | 0        |
| mmerzbank                              | 1            | 0         | 0         | 0      | 0       | 0       | 0       | 0        |
| morganstanley                          | 193          | 46        | 41        | 55     | 31      | 25      | 14      | 50       |
| nomura                                 | 78           | 15        | 19        | 50     | 7       | 2       | 2       | 14       |
| royalbankscotland                      | 109          | 22        | 17        | 52     | 24      | 7       | 7       | 12       |
| societegenerale                        | 98           | 19        | 11        | 83     | 4       | 0       | 1       | 6        |
| standardcharteredbank                  | 1            | 0         | 1         | 0      | 0       | 0       | 0       | 1        |
| suisse                                 | 192          | 45        | 42        | 75     | 31      | 17      | 13      | 41       |
| thehongkongandshanghaibankingrporation | 2            | 1         | 0         | 2      | 0       | 0       | 0       | 0        |
| ubs                                    | 136          | 32        | 31        | 71     | 22      | 7       | 4       | 25       |

*Replicated version*

Figure 10: Table OS3

- Table OS3: estimates do not match

| Variable      | Price after 30 Days | Price after 5 Days | Price after 1 Day |
|---------------|---------------------|--------------------|-------------------|
| IMM price     | 1.10<br>(0.40)      | 0.94<br>(0.34)     | 0.91<br>(0.29)    |
| Auction price | -0.08<br>(0.38)     | 0.07<br>(0.32)     | 0.11<br>(0.27)    |

*Paper version*

Table 9: Post-Auction Prices

|               | Price after 30 Days | Price after 5 Days | Price after 1 Day |
|---------------|---------------------|--------------------|-------------------|
| Auction price | -0.09<br>(0.38)     | 0.26<br>(0.32)     | 0.33<br>(0.27)    |
| IMM price     | 1.10<br>(0.40)      | 0.75<br>(0.33)     | 0.69<br>(0.28)    |
| Constant      | 1.38<br>(2.62)      | 1.95<br>(2.18)     | 0.73<br>(1.83)    |

$N = 62$ . Standard errors in parentheses.

*Replicated version*

Figure 11: Table OS3

- Table OS4: estimates do not match

TABLE OS.4. Bond Traits: Bidder Level

from nonparameteric regression on IMM and NOI. Bonds useful in  
t in bids conditional on IMM common signal and NOI. Standard error

| Variable      | IMM Submission  | Residualized bid |
|---------------|-----------------|------------------|
| Duration      | 0.08<br>(0.07)  | 0.09<br>(0.07)   |
| Conversion    | 14.89<br>(2.45) | 3.29<br>(1.59)   |
| Convexity     | -0.61<br>(0.56) | 0.49<br>(0.47)   |
| Volume        | 1.29<br>(1.18)  | -0.04<br>(0.72)  |
| Global Dealer | -3.06<br>(2.70) | -1.01<br>(1.08)  |
| Auction NOI   | Yes             | No               |
| Constant      | Yes             | Yes              |

| Table 10: Bond Traits: Bidder Level |                 |                  |
|-------------------------------------|-----------------|------------------|
|                                     | IMM Submission  | Residualized bid |
| Duration                            | 0.05<br>(0.07)  | 0.11<br>(0.07)   |
| Conversion                          | 11.56<br>(2.56) | 2.78<br>(1.63)   |
| Convexity                           | -3.04<br>(0.62) | -1.43<br>(0.53)  |
| Volume                              | 1.87<br>(1.18)  | 0.44<br>(0.71)   |
| Global Dealer                       | -0.25<br>(1.53) | -0.91<br>(1.11)  |
| $N$                                 | 2172            | 1165             |
| $R^2$                               | 0.075           | 0.014            |

Standard errors in parentheses. Controls: NOI, constant (col. 1); constant (col. 2).

*Replicated version*

Figure 12: Table OS4

- Table OS5: estimates do not match

TABLE OS.5. Bond Traits: Auction Level

1 influence of bond traits. Bond volume in billions. Standard e

| VARIABLES               | (1)<br>Auction Price | (2)<br>Auction IMM |
|-------------------------|----------------------|--------------------|
| duration                | 0.02<br>(0.03)       | 0.35<br>(0.25)     |
| conversion              | 2.71<br>(1.66)       | 42.38<br>(12.82)   |
| convexity               | 0.11<br>(0.27)       | 3.20<br>(2.11)     |
| volume                  | 0.001<br>(0.003)     | 0.001<br>(0.003)   |
| <i>NOI</i>              | -0.01<br>(0.002)     | -0.05<br>(0.01)    |
| <i>IMM</i>              | 0.96<br>(0.04)       |                    |
| <i>IMM</i> <sup>2</sup> | 0.0003<br>(0.0004)   |                    |
| Constant                | -0.94<br>(1.45)      | 50.05<br>(6.775)   |

Paper version



Table 11: Bond Traits: Auction Level

| VARIABLES        | (1) Auction Price  | (2) Auction IMM   |
|------------------|--------------------|-------------------|
| Duration         | 0.018<br>(0.034)   | 0.37<br>(0.27)    |
| Conversion       | 2.724<br>(2.008)   | 43.41<br>(15.74)  |
| Convexity        | 0.090<br>(0.304)   | 1.56<br>(2.44)    |
| Volume           | 0.000<br>(0.000)   | 0.00<br>(0.00)    |
| NOI              | -0.009<br>(0.002)  | -0.06<br>(0.02)   |
| IMM              | 0.96<br>(0.04)     |                   |
| IMM <sup>2</sup> | 0.0003<br>(0.0004) |                   |
| Constant         | -1.09<br>(1.72)    | 12.81<br>(13.235) |
| $N$              | 198                | 198               |
| $R^2$            | 0.987              | 0.144             |

Standard errors in parentheses.

*Replicated version*

Figure 13: Table OS5

- Table OS6: estimates do not match

TABLE OS.6. Change in Auction Format

across the designs so that we hold fixed the incentives to manipulate, the price limit is  $\pm 2$  cents and  $\pm 4$  cents.

|                                                    |  | Double Auction |               |
|----------------------------------------------------|--|----------------|---------------|
|                                                    |  | Supply=500     | Supply=100    |
| Mean Price $\mathbb{E}[\boldsymbol{P} P^M = 34.5]$ |  | [35.85,36.22]  | [36.88,36.95] |
| Std ( $\boldsymbol{P} P^M = 34.5$ )                |  | [0.49,0.72]    | [0.17,0.67]   |

Paper version

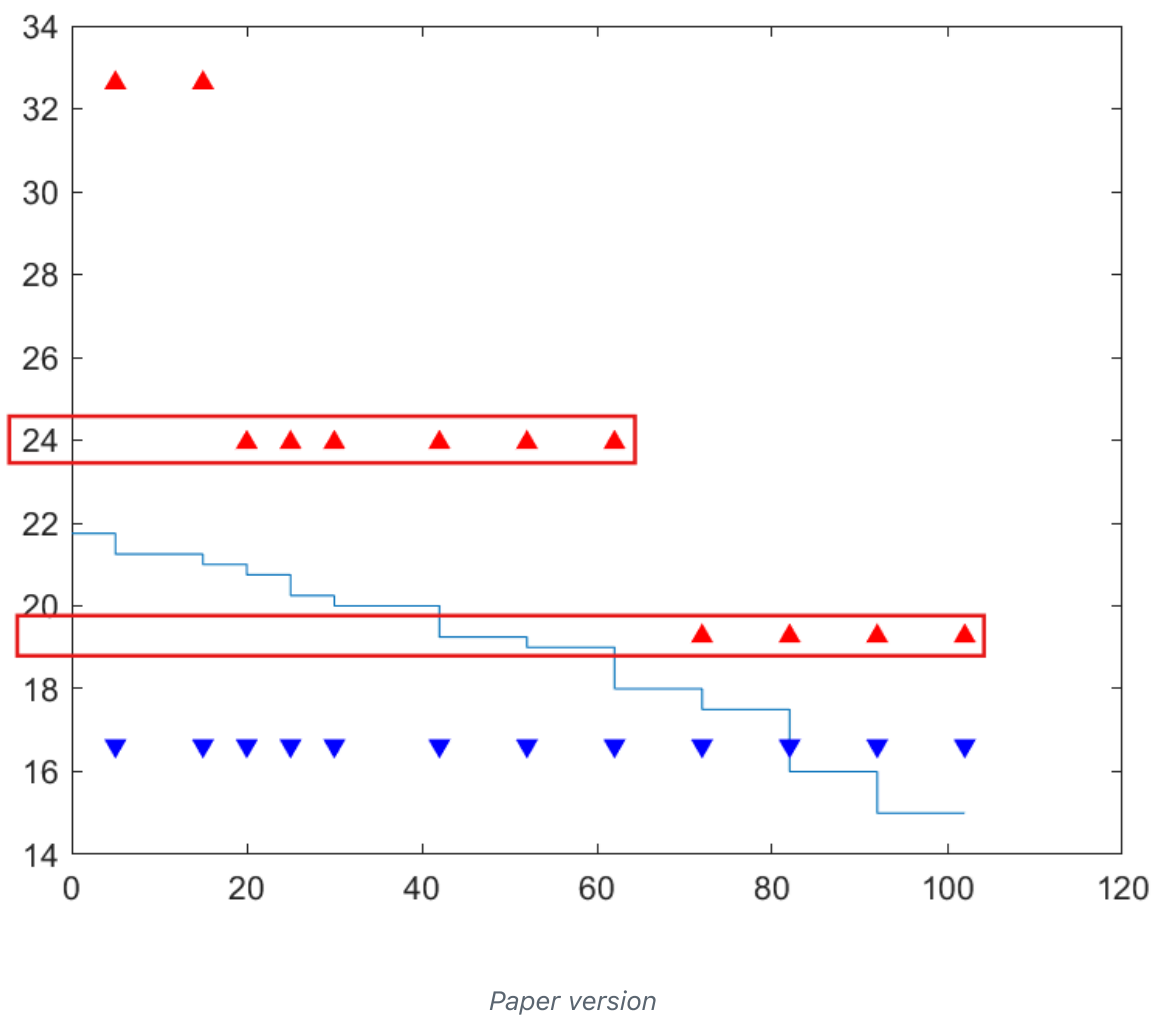
Table 12: Change in Auction Format (Bond Supply)

|                  | Mean Price     | SD Price     | Surplus (\$M) |
|------------------|----------------|--------------|---------------|
| Volume cap = 500 | [35.99, 36.27] | [0.47, 0.74] | [−1.6, 1.7]   |
| Volume cap = 100 | [36.87, 36.88] | [0.30, 0.30] | [−1.0, 1.5]   |

Replicated version

Figure 14: Table OS6

- Figure OS6 (left): output does not match



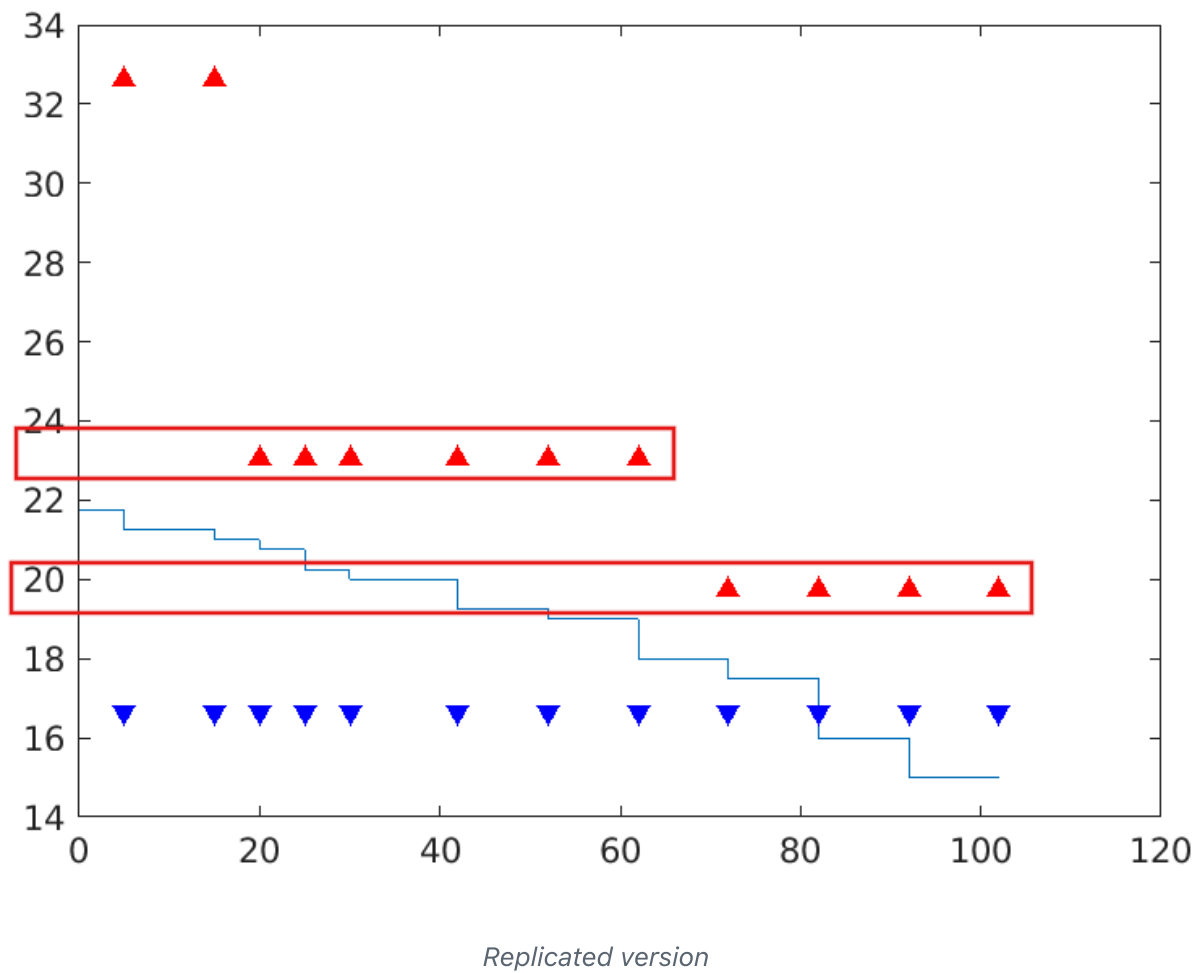
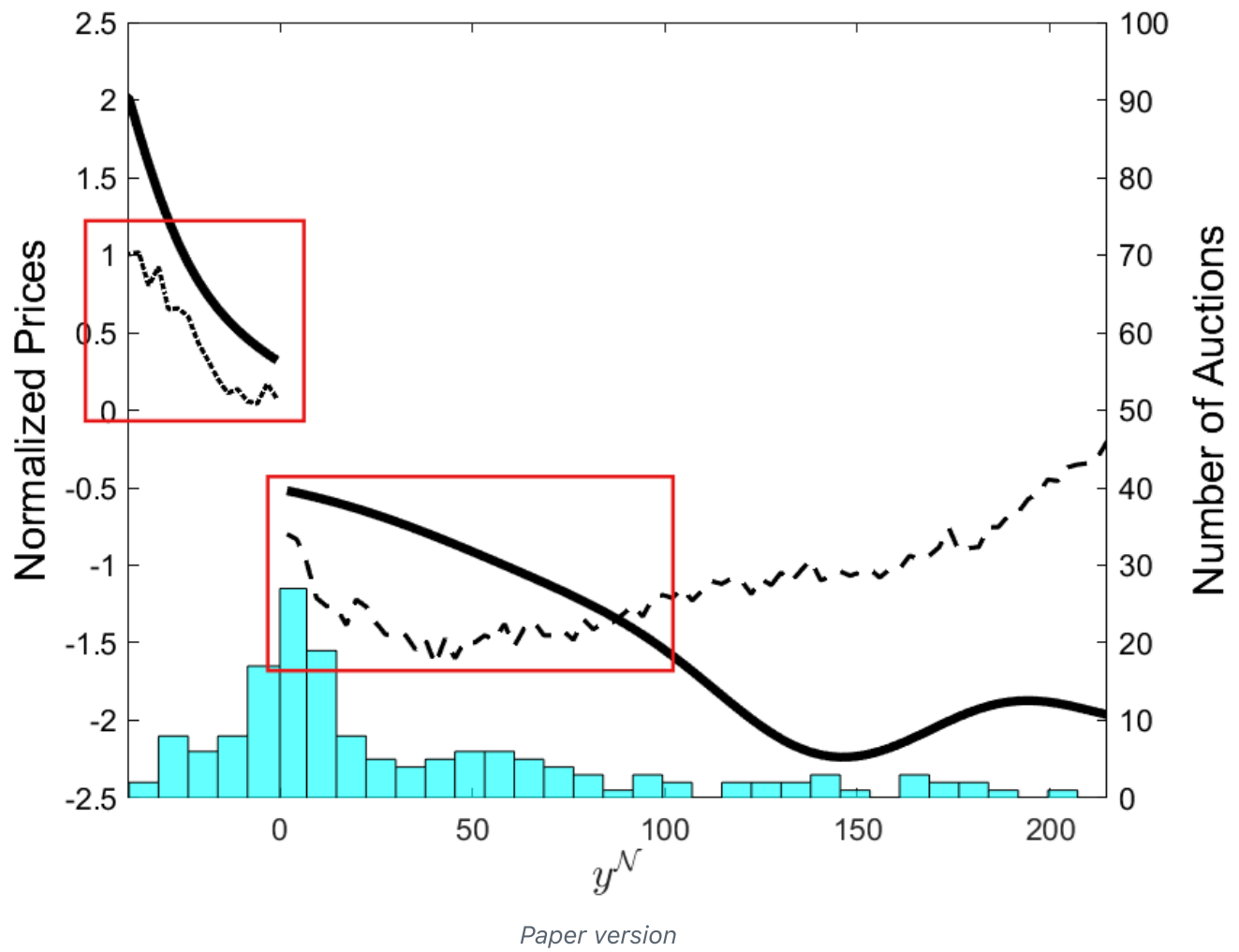


Figure 15: Figure OS6 (left)

- Figure OS8: output does not match



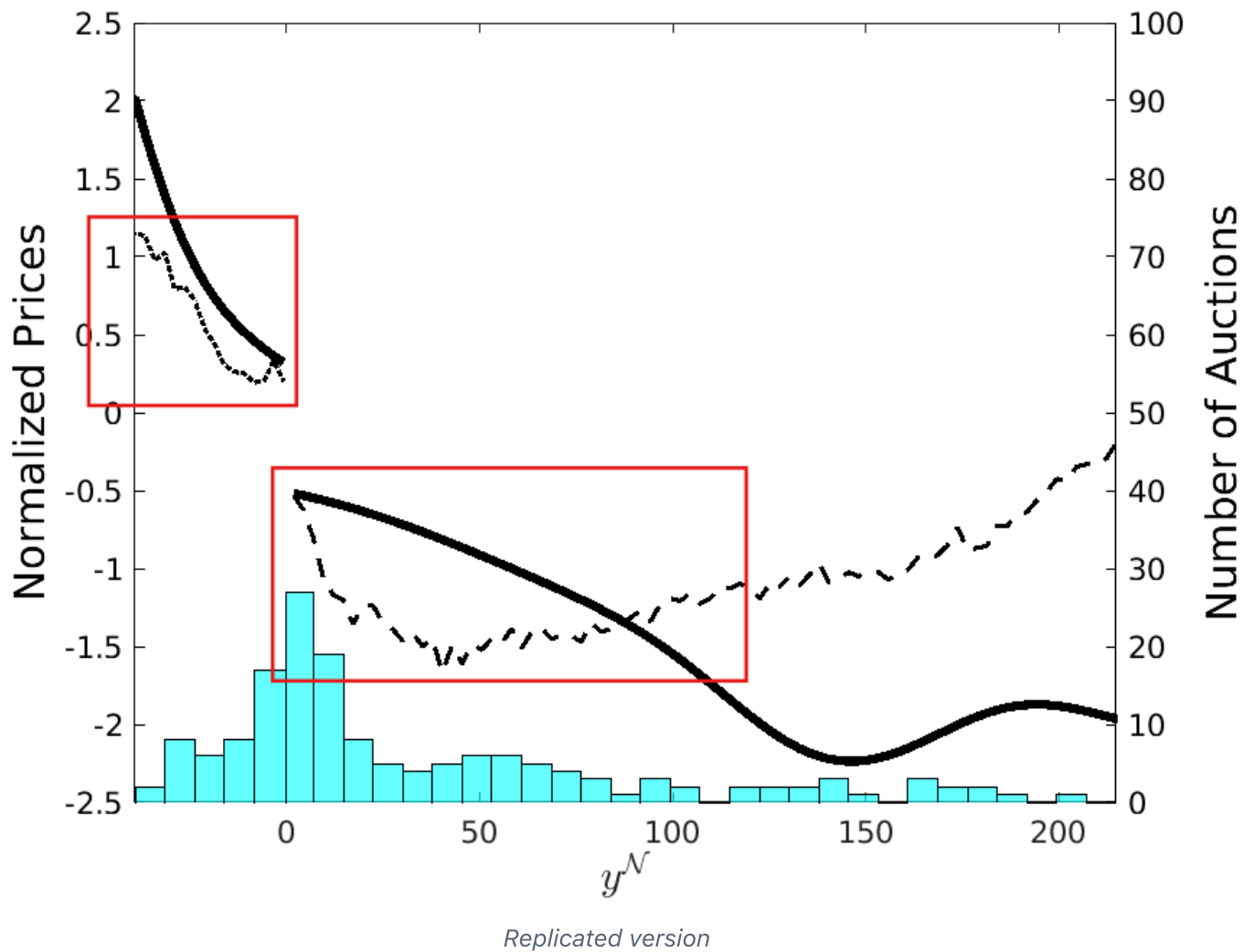


Figure 16: Figure OS8

## 9.3 In-Text Numbers

| In-text numbers                                                    | Program                | Line Number | Reproduced? |
|--------------------------------------------------------------------|------------------------|-------------|-------------|
| Section 2   Sample descriptives (N auctions, N bidders, NOI, etc.) | normalize_prices.m     |             | No          |
| Section 2   Price cap statistic                                    | firststagebidding.m    |             | Yes         |
| Section 6.1   Status quo surplus bounds                            | calculate_surplusraw.m |             | No          |

| In-text numbers                                     | Program                    | Line Number | Reproduced? |
|-----------------------------------------------------|----------------------------|-------------|-------------|
| Section 6.1   Truthful bidding surplus bounds       | postestimation_clean.m     |             | No          |
| Section 6.2   Price bias (cents, %)                 | postestimation_clean.m     |             | Yes         |
| Section 6.3   Risk SD, hedging effectiveness        | postestimation_clean.m     |             | No          |
| Section 7   CF surplus, price bias, hedging         | postmain_cfs.m             |             | No          |
| Appendix C.2   Risk aversion comparison             | riskaversion.m             |             | No          |
| Appendix C.5.1   Truthful reporting incentives      | truthfulpimmcheck.m        |             | No          |
| Appendix C.5.1   Quote manipulation calibration     | round1_quotescalibration.m |             | No          |
| Appendix C.6   Customer orders robustness           | robustnesschecks.m         |             | Yes         |
| Appendix C.7   Bid shading with positions           | postestimation_clean.m     |             | No          |
| Appendix D   CF robustness (volume caps, positions) | smc_cfs_yin.m              |             | No          |

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code
- Identification in README is ambiguous. Some results are in `cdsresults_log2.txt` and others in `cdsresults_log.txt`. It would be optimal to specify which ones are in which file.
- For Section 2, the log prints don't match one of the in-text numbers:
  - "Of these, 84 were loan credit default swaps (LCDS) and 114 CDS"

- "The cap plays an important role, with 15% of auctions clearing at the cap (6% conditional on non-zero excess supply/demand)."

===== SECTION 2: Sample Composition =====

Total credit events: 202

LCDS auctions: 47

CDS auctions: 155

Fraction of bids at price floor/ceiling: 3.1%

Fraction of auctions clearing at cap: 15.2%

Fraction at cap conditional on nonzero NOI: 16.8%

- For Section 6.1, the log prints don't match one of the in-text numbers:
  - "The surplus of the status-quo design is \$[-7, 48]M".

Status quo surplus bounds: [\$-8M, \$47M]

===== SECTION 6.1: Surplus from Reallocation =====

Truthful bidding surplus bounds: [\$43M, \$91M]

- For Section 6.3, the log prints don't match in-text numbers:
  - "yields a standard deviation of wealth of 0.77 cents."
  - "with standard deviation of 1.1 cents"
  - "by [84.5,87.9]%."
  - "covers [87.5,91.7]% of the risk"
  - "bond owners would hedge [75.8, 99.1]%"

===== SECTION 6.3: Risk from Price Uncertainty =====

SD of risk (lower bound): 0.75 cents

Optimal hedging range: [75.6%, 99.6%]

Optimal hedging position: 91.5%

Hedging effectiveness at n=1 (SD): [84.0%, 88.0%]

Hedging effectiveness at optimal (SD): [87.4%, 91.9%]

Hedging effectiveness at n=1 (variance): 98.4%

Hedging effectiveness at optimal (variance): 99.3%

- For Section 6.4, the log prints in-text have not been mentioned in the table of the [README.md](#).  
Below the output:

===== SECTION 6.4: CDS Market Impact =====

Max insurance coverage: 91.9%

Auction bias charge (basis points): 1.9 bps

Firm value gain from full insurance (Danis-Gamba scaling):  $0.26\% = 2.9/91.9 * (100 - 91.9)/100$

Dollar gain per average firm (\$M): 132

IdleTimeout has been reached.

Parallel pool using the 'Processes' profile is shutting down.

Repurchase rate (conditional on possible): 61.1%

However, numbers don't match the text: "(so that the 2.9% increase in firm value is 91.7% of the benefit from a full insurance contract), a CDS contract that provided full insurance could result in



value gains of 3.16%(= 2.9 0.917 ), resulting in firm value gains of 0.26%, or \$130 million for the average firm."

- For Appendix C.2, "for a range of CARA utility coefficients from 0.005 to 10". However, the log shows:

```
--- Appendix C.2: Risk Aversion Check ---
Variance of baseline wealth: 0.008219
Variance of wealth2 (auction outcomes): 0.002546
Variance of wealth2 (H/T doubling): 0.000402
Variance ratio (baseline/H-T auction gamble): 20.4x
Utility ratio range [min, max]: [0.9951, 0.9997]
```

- For Appendix C.5.1, "[...] this gives an increase of 1756 dollars of surplus." In the log we have:

```
===== ROBUSTNESS CHECKS =====
--- Appendix C.5.1: Truthful Reporting Incentives ---
P95 benefit of manipulation: 1762.50
Mean cost of manipulation: -40214
dR/dp_quote (avg effect on IMM): 0.5548
--- Appendix C.5.1: Quote Manipulation Calibration ---
Calibrated parameters: sigma_eta=0.55, mpb=-10.0, sigpb=1.0
Variance ratio V(E[R|IMM])/V(E[R|eta]): 0.0020
```

- For Appendix C.7, "At the 10th percentile (across bidders) the 95% CI for shading is [-8.00,0.04] cents, the median [-1.42,1.58] cents, and at the 90th percentile [-0.01,7.50] cents". However, the log shows:

```
===== APPENDIX C.7: Bid Shading from Positions =====
Mean absolute shading [LB, UB]: [2.564, 2.670]
Median shading [LB, UB]: [-1.196, 1.478]
Shading percentiles (LB): P10=-8.000 P25=-3.317 P50=-1.196 P75=-0.279 P90=-0.011
Shading percentiles (UB): P10=0.025 P25=0.302 P50=1.478 P75=3.485 P90=7.596
```

- For Appendix D, the log print does not provide the in-text numbers

## 9.4 Other Issues

- Missing storage requirement. 56 GB are required for the intermediate files.
- Figure OS.3 output is split in two images. The second `abnormaleventgraph_mean.png` is not mentioned in the `README.md` output table

## 10 Classification

- ☐ full reproduction
- ☒ full reproduction with minor issues
- ☐ partial reproduction (see above)

- ☐ not able to reproduce most or all of the results (reasons see above)

## 10.1 Reason for incomplete reproducibility

---

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
  - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements